

Static Detection of Second-Order Vulnerabilities in Web Applications

Static Detection of Second-Order Vulnerabilities in Web Applications - Author not stated, [usenix.org](https://www.usenix.org).

- Published: date not stated
- Original: <<https://www.usenix.org/conference/usenixsecurity14/technical-sessions/presentation/dahse>>
- Preserved from: <https://www.usenix.org/conference/usenixsecurity14/technical-sessions/presentation/dahse> (live) on 2026-08-08
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Friday, August 1, 2014 - 10:45am

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Web applications evolved in the last decades from simple scripts to multi-functional applications. Such complex web applications are prone to different types of security vulnerabilities that lead to data leakage or a compromise of the underlying web server. So called *secondorder vulnerabilities* occur when an attack payload is first stored by the application on the web server and then later on used in a security-critical operation.

In this paper, we introduce the first automated static code analysis approach to detect second-order vulnerabilities and related multi-step exploits in web applications. By analyzing reads and writes to memory locations of the web server, we are able to identify unsanitized data flows by connecting input and output points of data in *persistent data stores* such as databases or session data. As a result, we identified 159 second-order vulnerabilities in six popular web applications such as the conference management systems *HotCRP* and *Open- Conf*. Moreover, the analysis of

web applications evaluated in related work revealed that we are able to detect several critical vulnerabilities previously missed.

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BibTeX

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@inproceedings {184419, author = {Johannes Dahse and Thorsten Holz}, title = {Static Detection of {Second-Order} Vulnerabilities in Web Applications}, booktitle = {23rd USENIX Security Symposium (USENIX Security 14)}, year = {2014}, isbn = {978-1-931971-15-7}, address = {San Diego, CA}, pages = {989--1003}, url = {https://www.usenix.org/conference/usenixsecurity14/technical-sessions/presentation/dahse}, publisher = {USENIX Association}, month = aug }
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